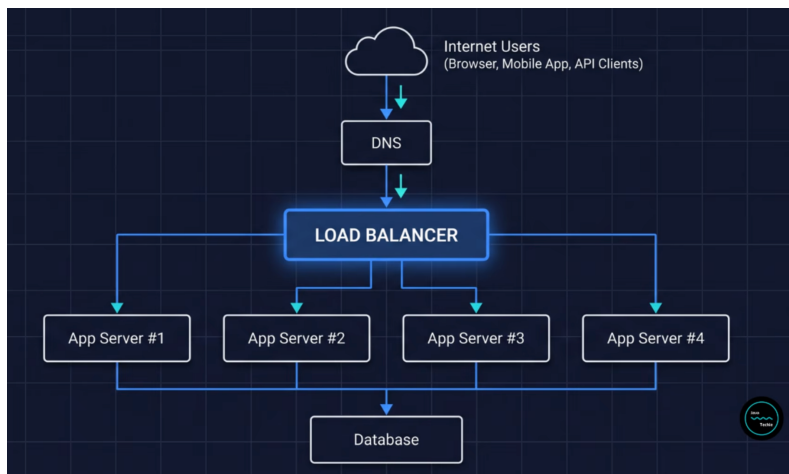
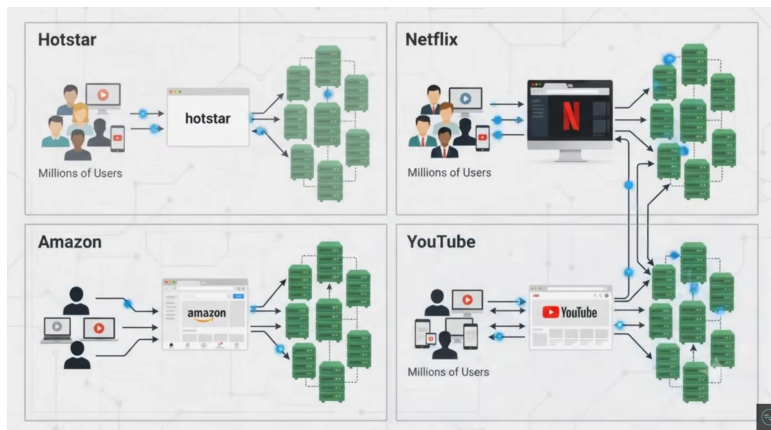
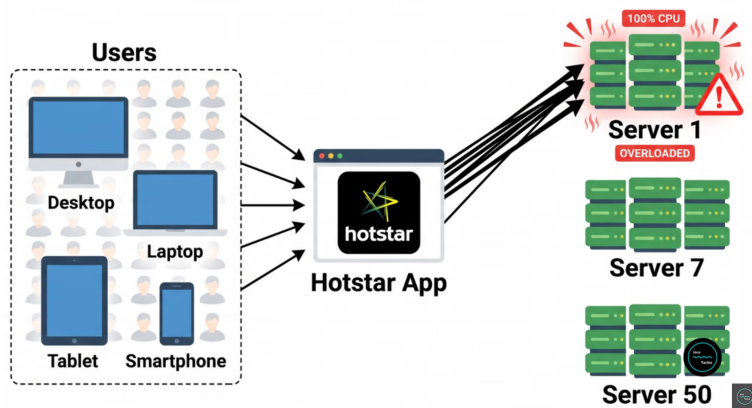
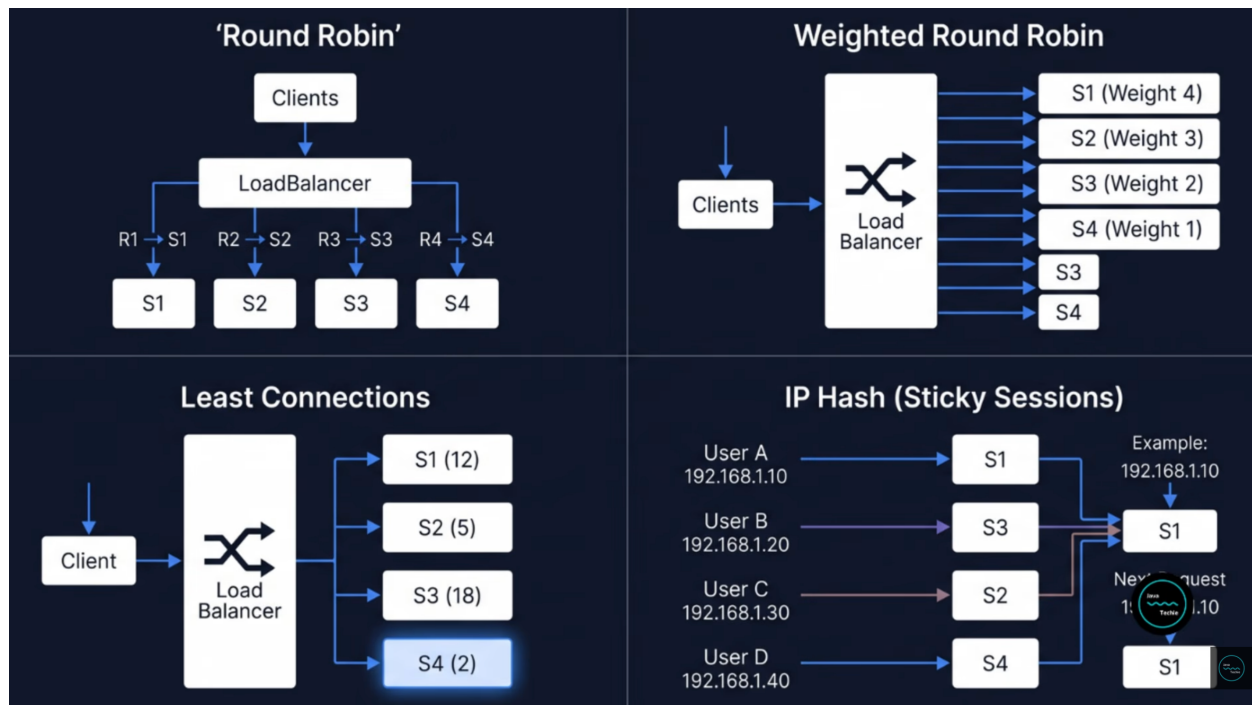


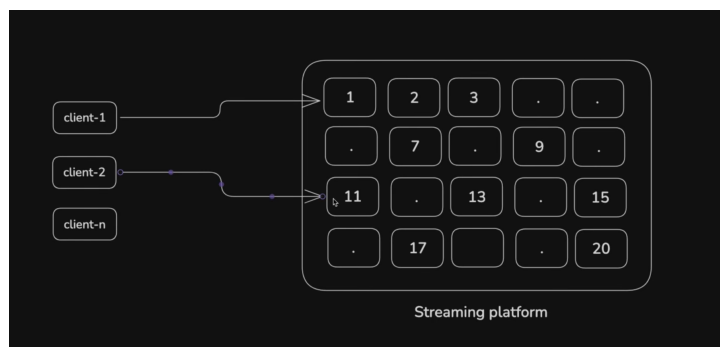
Load Balancer



Load balance Technic

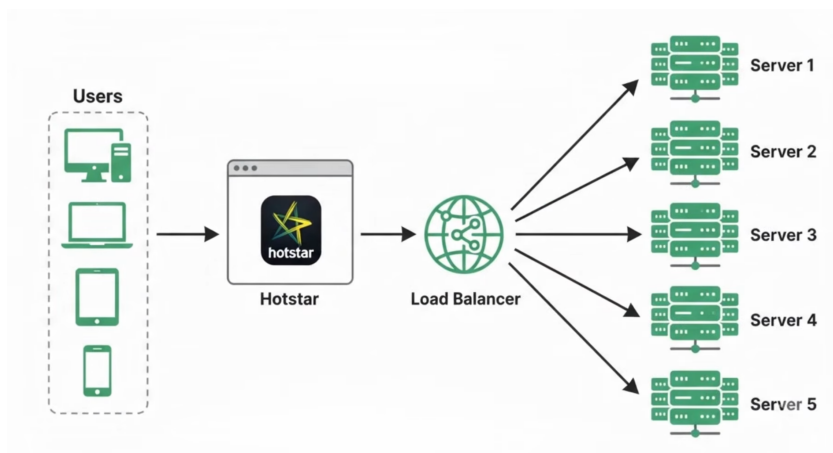


Who will decide which client request should go to which server out of 20 servers

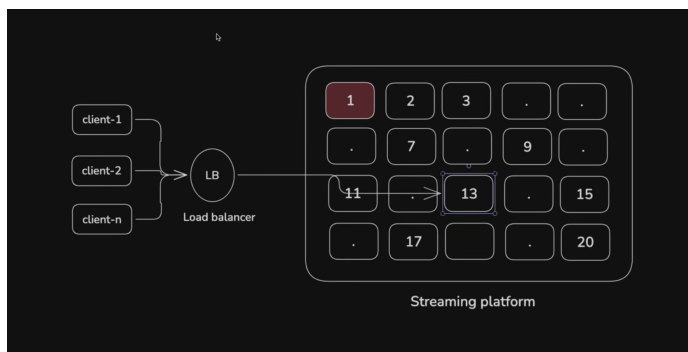


LD does traffic police job





Load balancer uses multiple factors which server is healthy/ available / the routing algorithm being used and identify the best one and route. Ensure no single server is overloaded.



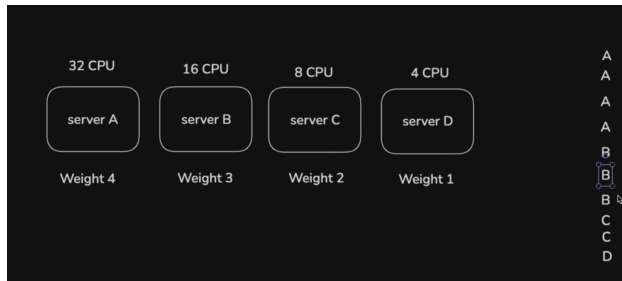
1. Round Robin
2. Weighted Round Robin
3. least Connections
4. IP Hash (Sticky Sessions)

Round robin-> requests direct in circular order to servers. Distribute traffic evenly under assumption every server has same capacity and configuration.

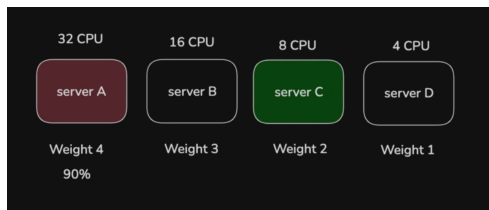


Weighted Round Robin -> what if servers have different capacity.

This algorithm does weight to each server. Load balancer distributes traffic based on weight. First 4 requests go to server A, next 3 requests go to server B, etc..



Not bother about load event server c is free as per weight LB route to Server A even its CPU usage is max



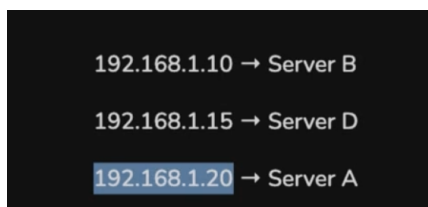
Least connection: which server has list active connection.



Same user can redirect different servers as result Session gap. Session available in Server A but not in Server C.

Then we come IP hashing. Every time request from same IP goes to same server.

This technic is called session affinity / sticky session. When session is important this technic is good.



Based on our use case choose the best.